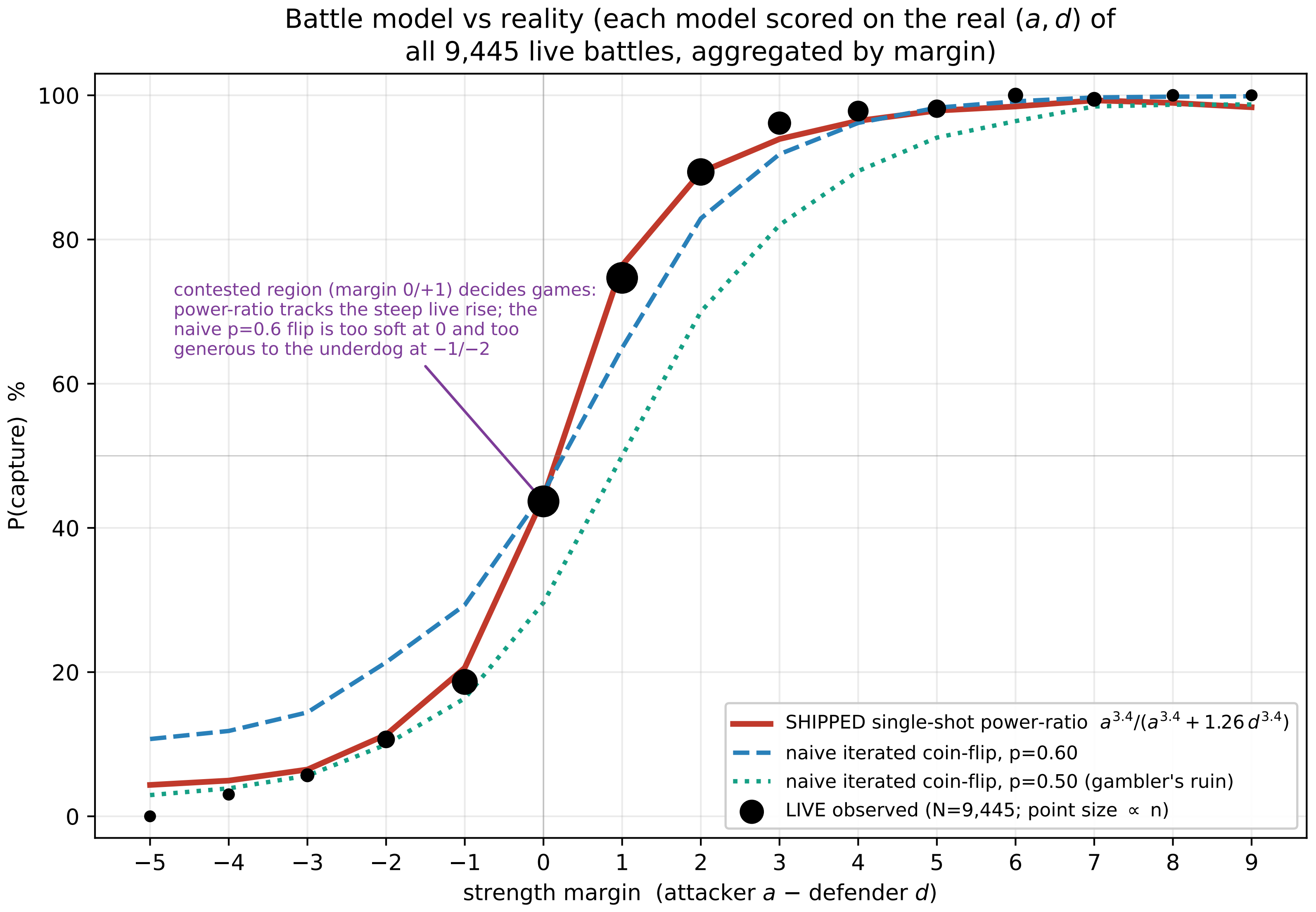
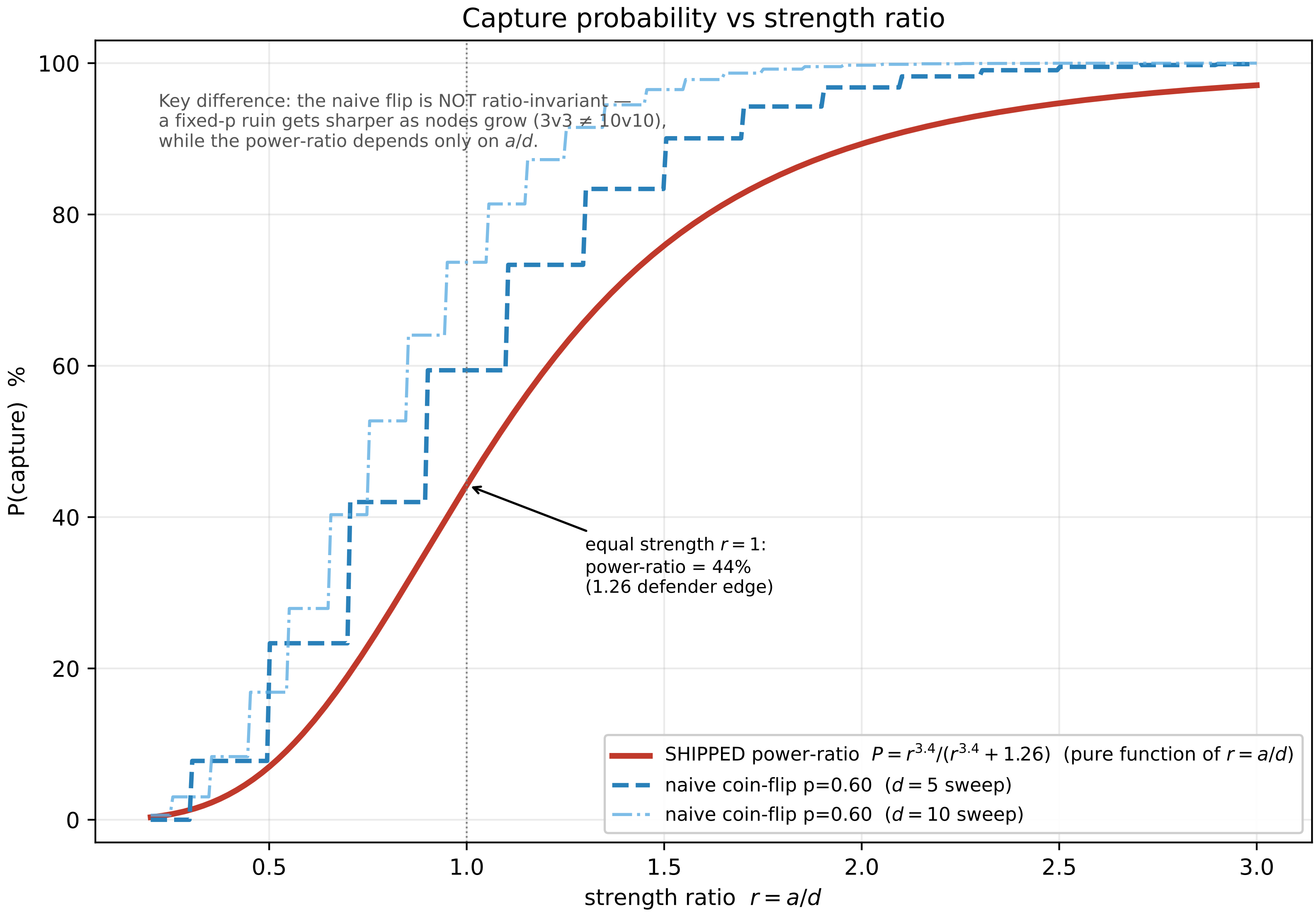


Battle model vs reality (each model scored on the real (a, d) of
all 9,445 live battles, aggregated by margin)
contested region (margin 0/+1) decides games:
power-ratio tracks the steep live rise; the
naive p=0.6 flip is too soft at 0 and too
generous to the underdog at −1/−2
P(capture) %
0
20
40
60
80
100
−5
−4
−3
−2
−1
0
1
2
3
4
5
6
7
8
9
strength margin (attacker a − defender d)
SHIPPED single-shot power-ratio $a^{3.4}/(a^{3.4} + 1.26\,d^{3.4})$
naive iterated coin-flip, p=0.60
naive iterated coin-flip, p=0.50 (gambler's ruin)
LIVE observed (N=9,445; point size ∝ n)



Capture probability vs strength ratio
Key difference: the naive flip is NOT ratio-invariant —
a fixed-p ruin gets sharper as nodes grow (3v3 ≠ 10v10),
while the power-ratio depends only on a/d.
equal strength $r = 1$:
power-ratio = 44%
(1.26 defender edge)
SHIPPED power-ratio $P = r^{3.4}/(r^{3.4} + 1.26)$ (pure function of $r = a/d$)
naive coin-flip p=0.60 ($d = 5$ sweep)
naive coin-flip p=0.60 ($d = 10$ sweep)
P(capture) %
strength ratio $r = a/d$
0
20
40
60
80
100
0.5
1.0
1.5
2.0
2.5
3.0

Fit to 9,445 live battles — models scored on real (a, d)

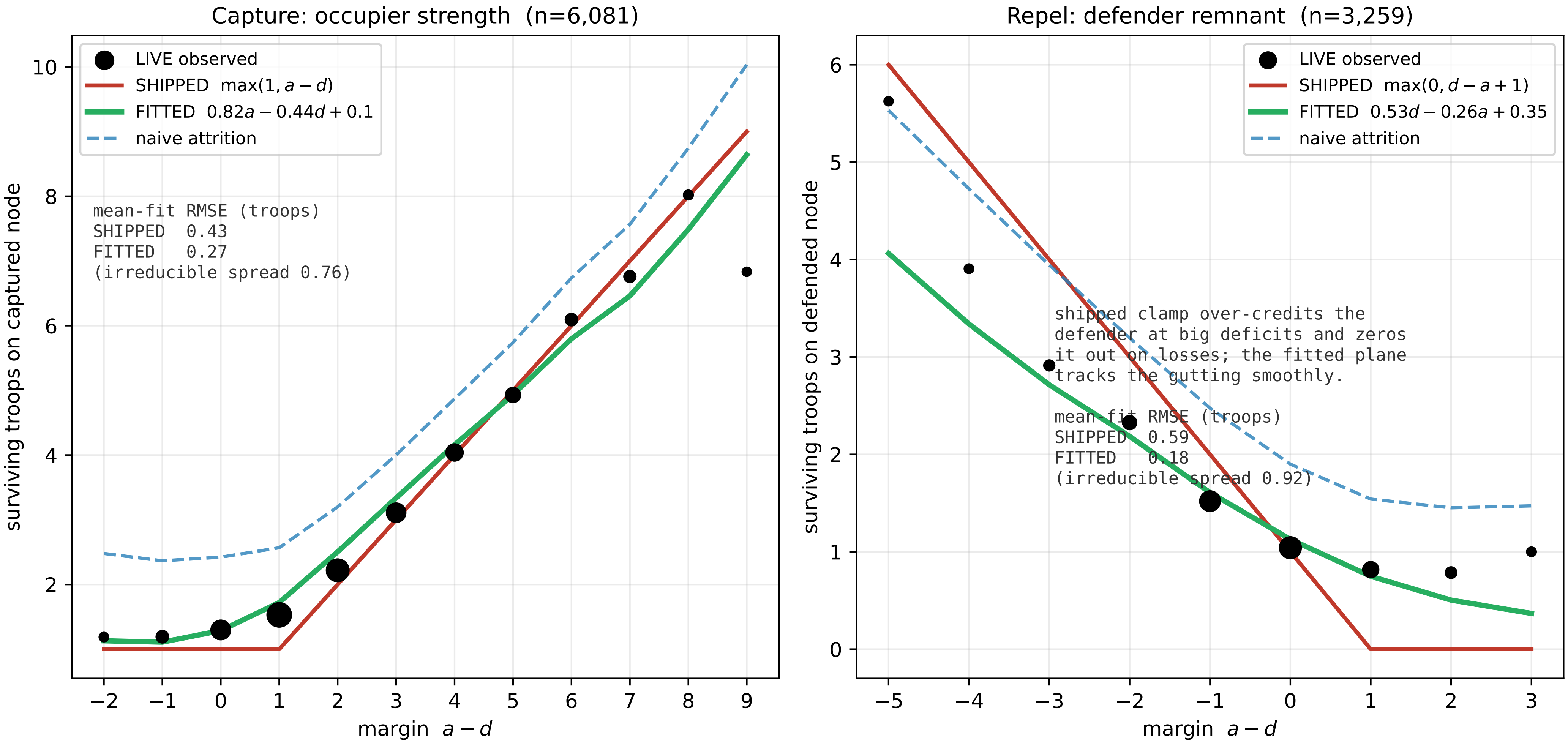
margin	n	live %	power-ratio %	|err|	naive 0.6 %	|err|
-3	123	5.7	6.5	0.8	14.4	8.7
-2	375	10.7	11.3	0.6	21.4	10.7
-1	1,175	18.6	20.6	2.0	29.3	10.7
+0	1,962	43.7	44.2	0.6	44.9	1.2
+1	1,939	74.7	76.4	1.7	65.0	9.7
+2	1,364	89.4	89.3	0.0	82.9	6.4
+3	881	96.1	93.9	2.2	91.8	4.3
+4	640	97.8	96.5	1.4	96.2	1.6
+5	431	98.1	97.9	0.3	98.3	0.1
+6	204	100.0	98.5	1.5	99.2	0.8
RMSE			**1.4**		**6.9**	

Per-battle goodness of fit (all 9,445 battles)

```
log-likelihood:  power-ratio -3,876  >  naive 0.6 -4,053  >  naive 0.5 -4,484   (higher better)
Brier score:      power-ratio 0.1324  <  naive 0.6 0.1397  <  naive 0.5 0.1570   (lower better)
```

Matches BATTLE_FUNCTION.md §6: single-shot power-ratio is both the simplest model (one closed-form Bernoulli, no loop) and the best fit. The naive fixed-p coin-flip is too soft at the contested margins.

Beyond win/loss: how many troops remain. Source node $\rightarrow 1$ in ~100% of battles. Fitted (a, d) planes beat the margin clamp.



Dropped 105 physically-impossible survivor reads (occupier $> a-1$ or remnant $> d$ — OCR/reinforcement artifacts that inflated the rare-outcome tails).